

Homework 3

ASEN 5090 Introduction to GNSS

Justin Le

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Problem 1: GPS Observations

(a.)

% Read rinex file`rinexData = rinexread('NIST00USA_R_20262310000_01D_30S_MO.rnx');``GPS_rinexData = rinexData.GPS;``PRN_number = 5;``PRN_index = GPS_rinexData.SatelliteID == PRN_number;``PRN05_GPS_rinexData = GPS_rinexData(PRN_index,:);``[NIST_GPS_week, NIST_GPS_TOW] = utc2gpstime(PRN05_GPS_rinexData.Time);``ephem_time_s = NIST_GPS_week.*604800 + NIST_GPS_TOW;`*% Nan gap the discontinuous*`gap_idx = [false; diff(ephem_time_s) > 1.5*median(diff(ephem_time_s))];``PRN05_GPS_rinexData.C1C(gap_idx) = NaN;``PRN05_GPS_rinexData.S1C(gap_idx) = NaN;``PRN05_GPS_rinexData.C1W(gap_idx) = NaN;``PRN05_GPS_rinexData.D1C(gap_idx) = NaN;`

(b.)

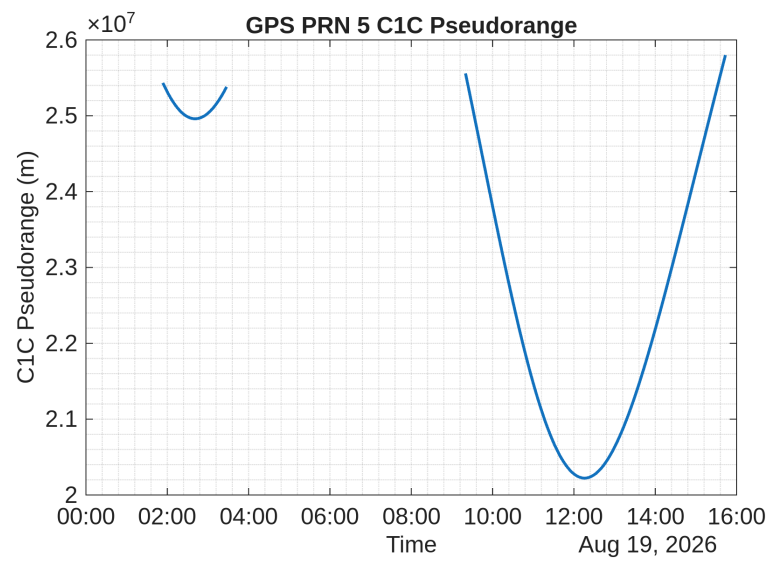


Figure 1: C1C Pseudorange for PRN5

(c.)

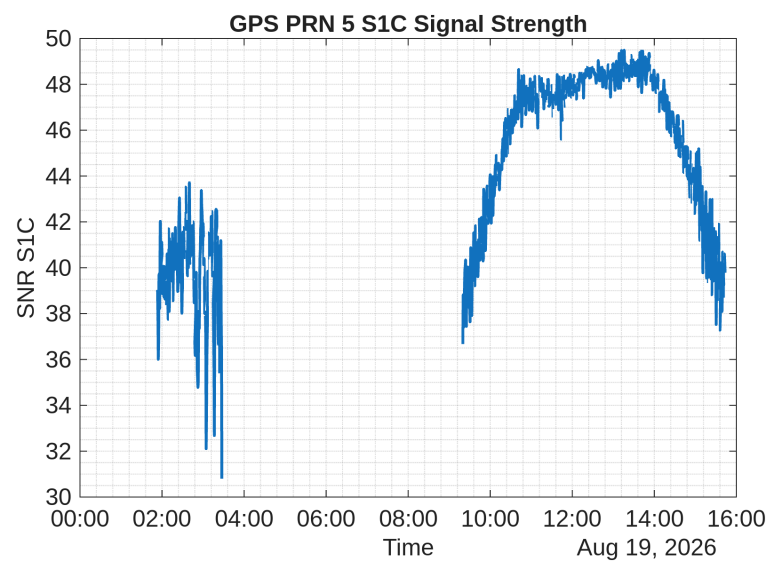


Figure 2: S1C SNR for PRN5

(d.)

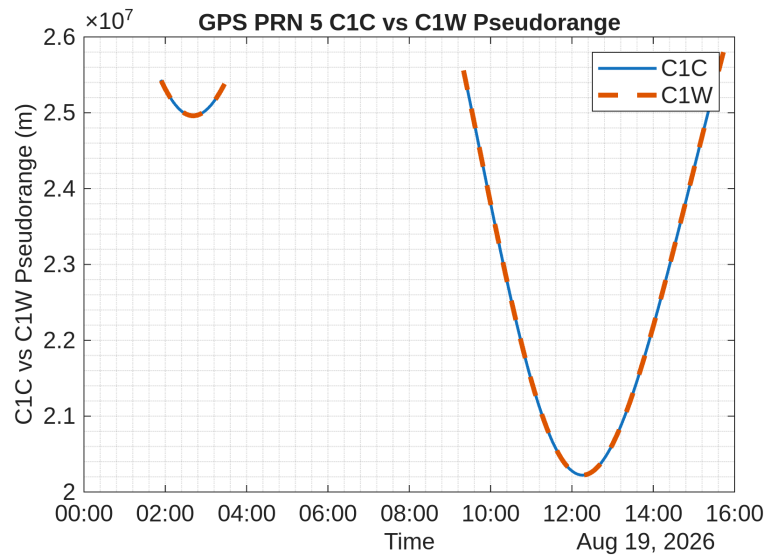


Figure 3: C1C and C1W Pseudorange for PRN5

TODO: describe the differences yada yada

Problem 2: GPS Positions Based on Ephemeris Data

(a.)

```
% Use read clean to read in ephermides
clean_GPSbroadcast = read_clean_GPSbroadcast('brdc2310.26n', true);

% Read in sp3 ephem like HW2
sp3 = read_sp3('IGS00PSFIN_20262310000_01D_15M_ORB.SP3');
satPRN = extract_sp3_prn(sp3);

[health,satPos,satVel,satClkCorr] = eph2pvt2025(clean_GPSbroadcast, [satPRN(5).
    week_number satPRN(5).TOW_s], 5);
```

(b.)

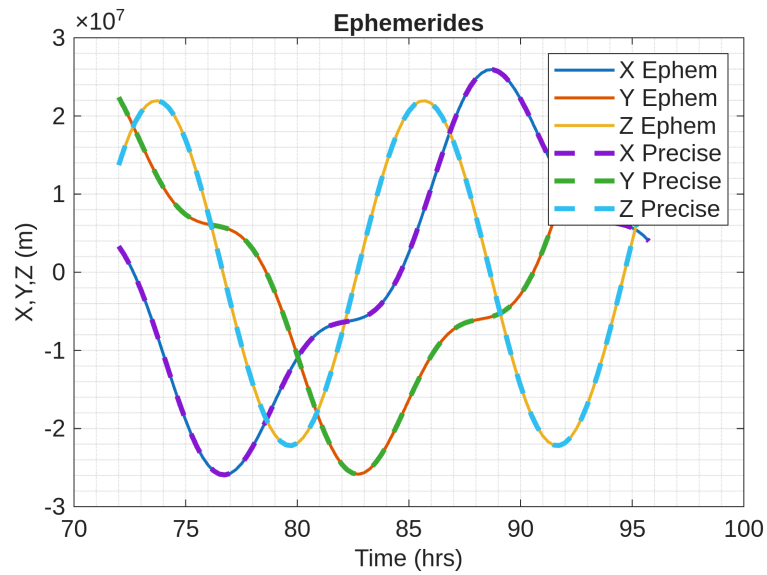


Figure 4: Ephemeris vs Precise

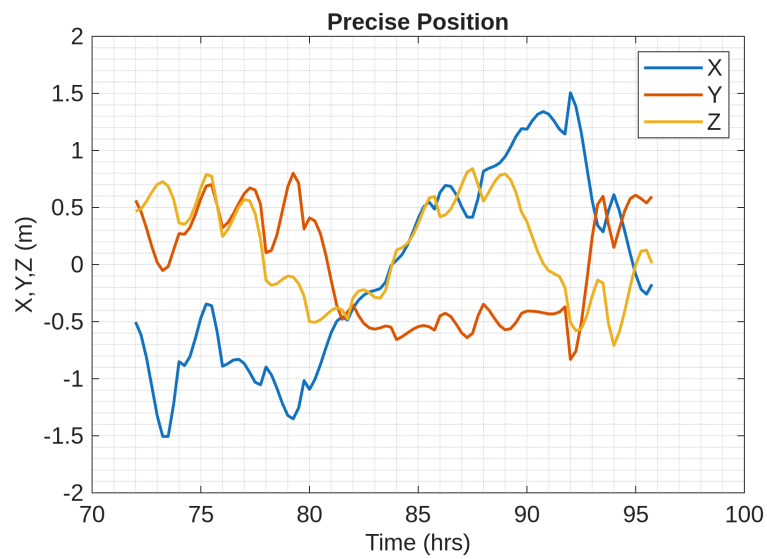


Figure 5: Ephemeris vs Precise

TODO: Comment on differences

(c.)

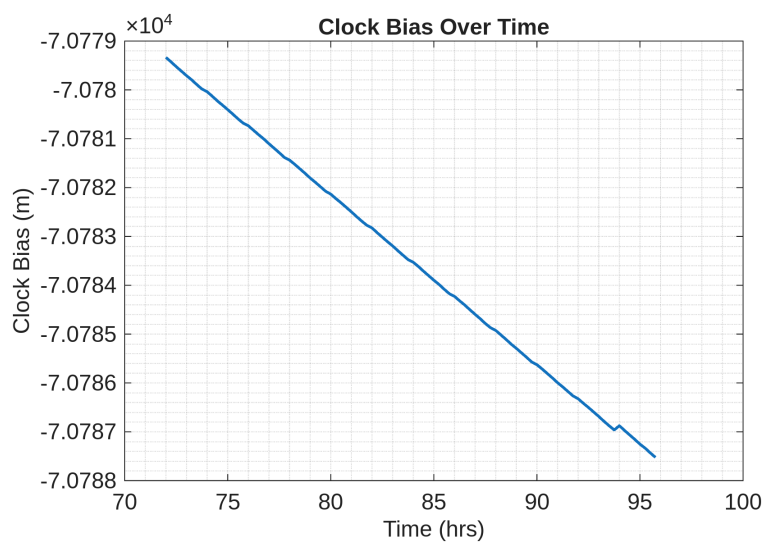


Figure 6: Clock Bias

Problem 3: GPS Expected Range

(a.)

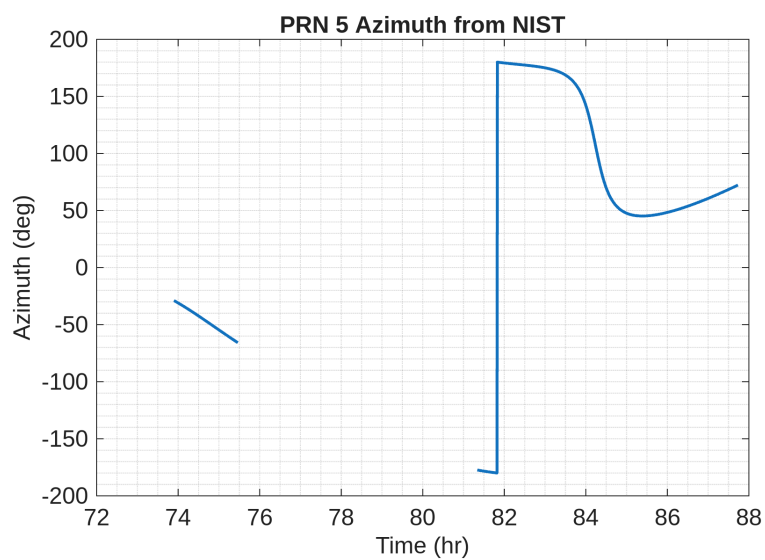


Figure 7: PRN5 Azimuth

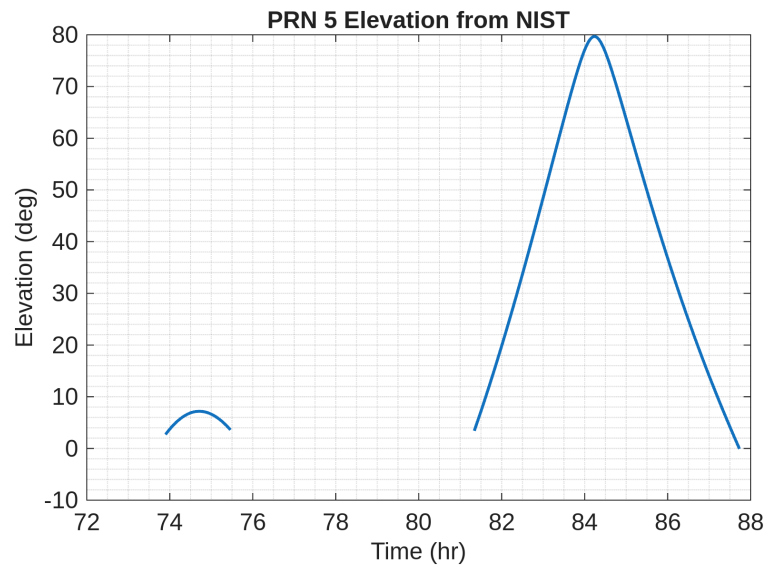


Figure 8: PRN5 Elevation

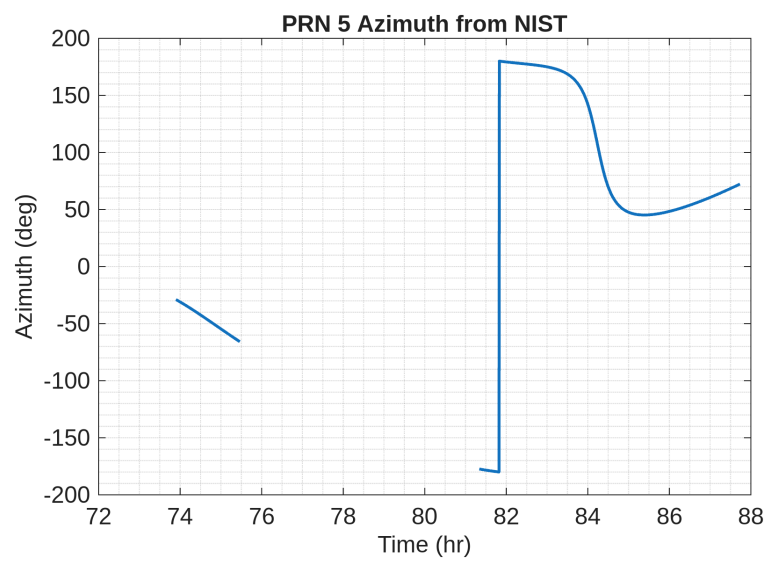


Figure 9: PRN5 Azimuth

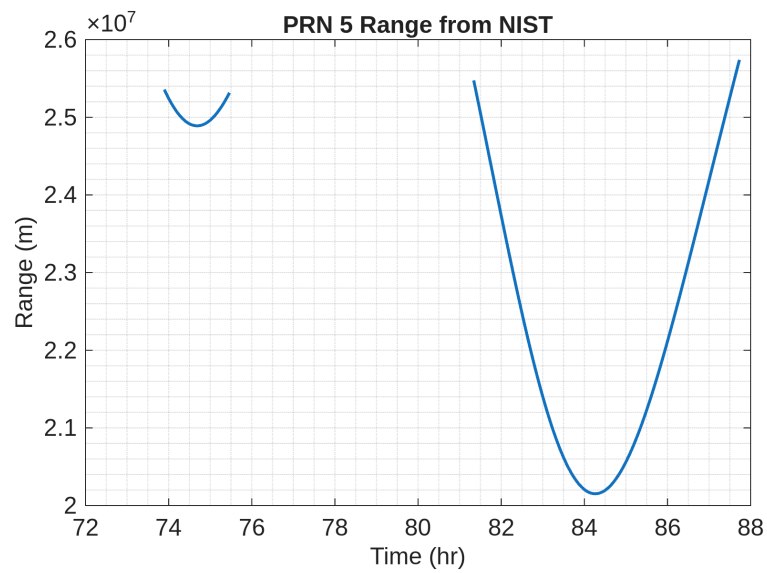


Figure 10: PRN5 Range

(b.)

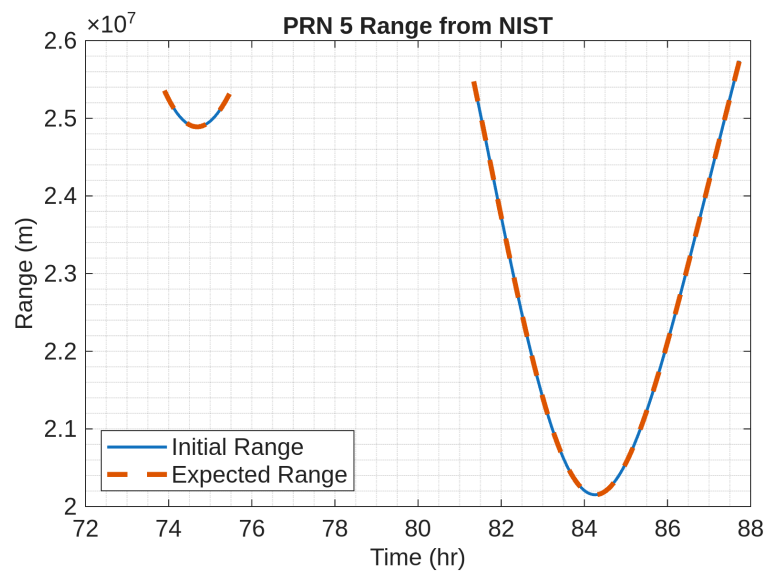


Figure 11: PRN5 Range Initial vs Expected

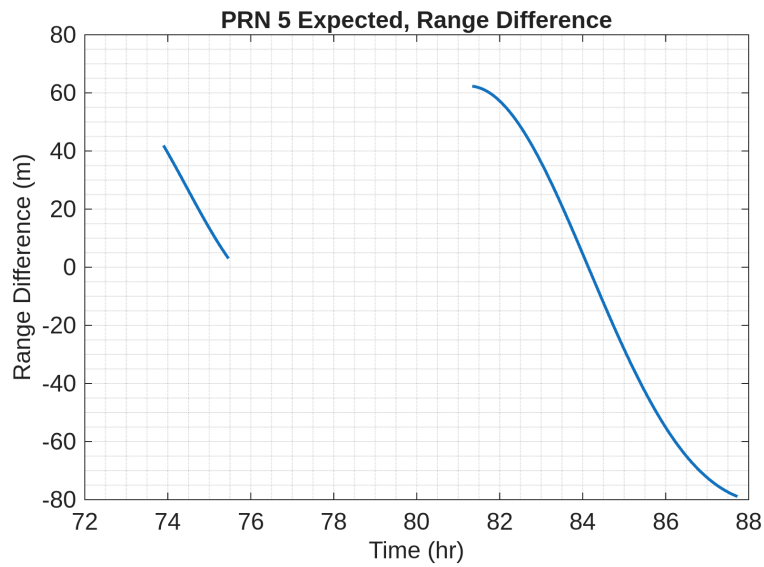


Figure 12: PRN5 Range Initial vs Expected Residual

TODO: report largest difference found.

Problem 4: Observed Minus Computed (Expected) Range

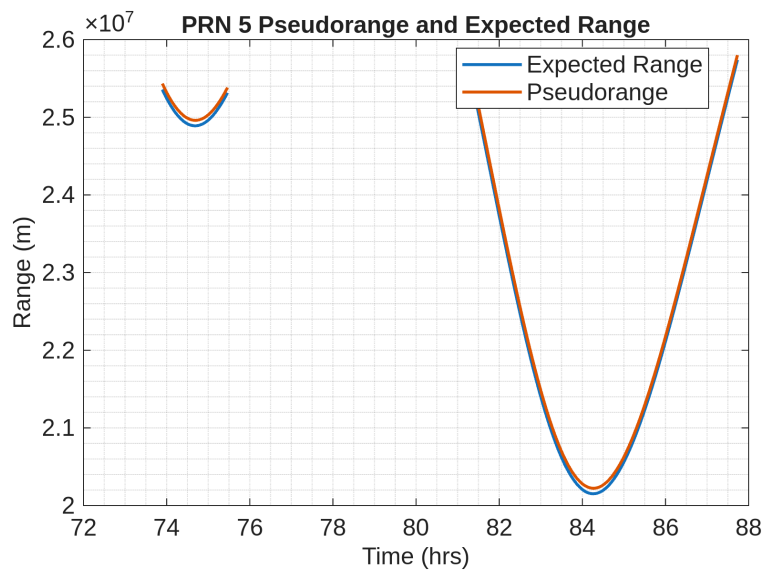


Figure 13: PRN5 C1C Pseudorange vs Expected Range

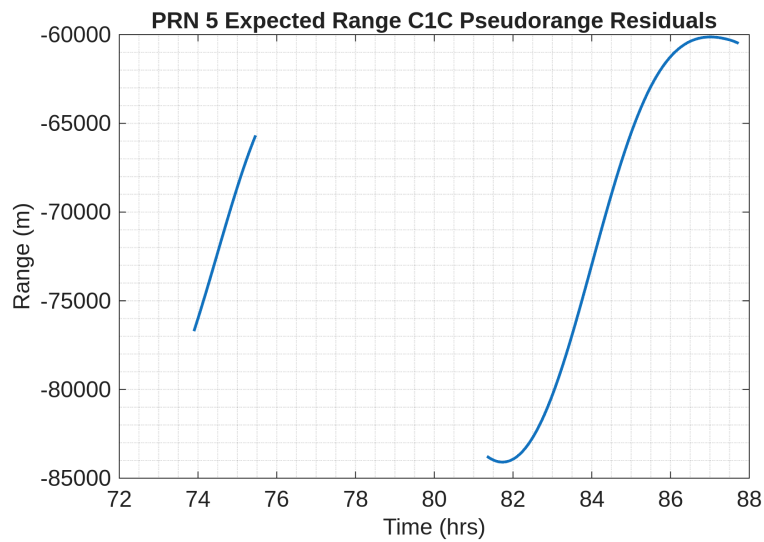


Figure 14: PRN5 C1C Pseudorange vs Expected Range Residuals

Problem 5: Repeat for another PRN

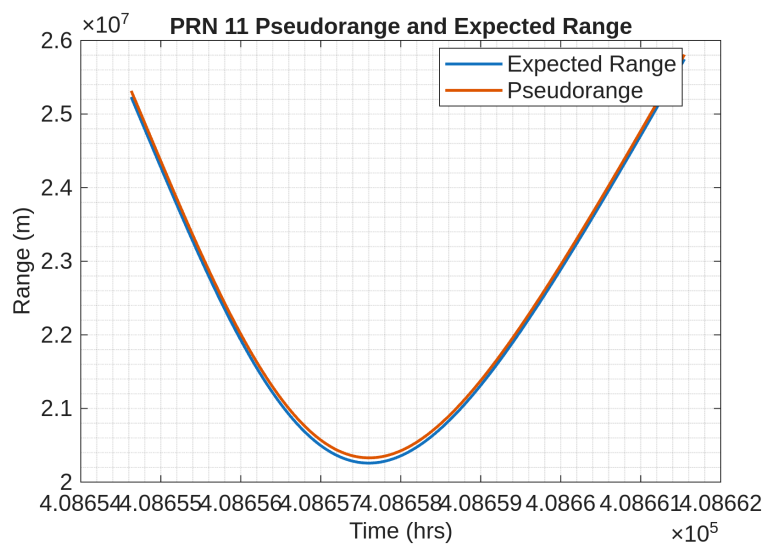


Figure 15: PRN11 C1C Pseudorange vs Expected Range

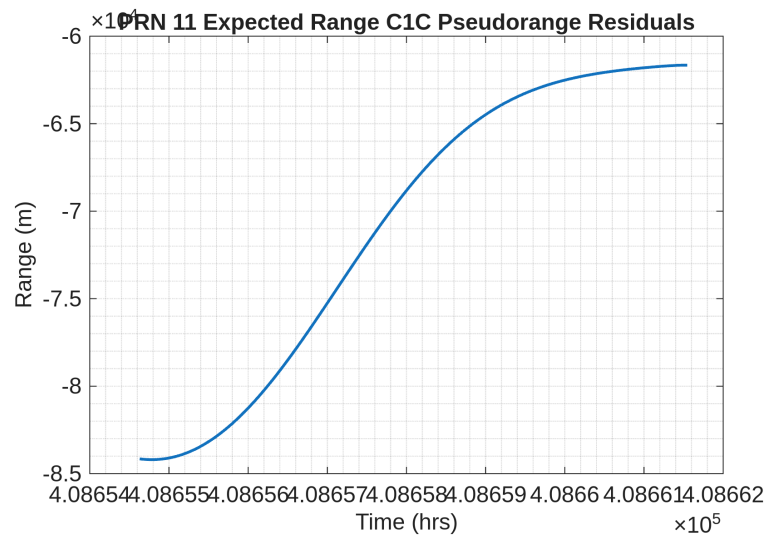


Figure 16: PRN11 C1C Pseudorange vs Expected Range Residuals